

SIA GOLDEN GREENS CROP | AI & AGTECH

GOLDEN GREENS CROP

The Cyber-Physical AI "Brain" for Smart Aquaponics &
Medicinal Herb CEA (Data Center Heat Integrated)

📍 RIGA, EUROPEAN UNION

STRATEGIC CONTEXT

THE THERMAL PARADOX

Why are we wasting energy in the digital age when agriculture is
starving for efficient heat?

| THE INDUSTRY PAIN POINTS



DATA CENTER LIABILITY

Global data centers reject 338 PJ of heat annually. Cooling these systems represents up to 40% of their operational energy costs—a pure financial and environmental waste.



AGRICULTURAL ENERGY CRISIS

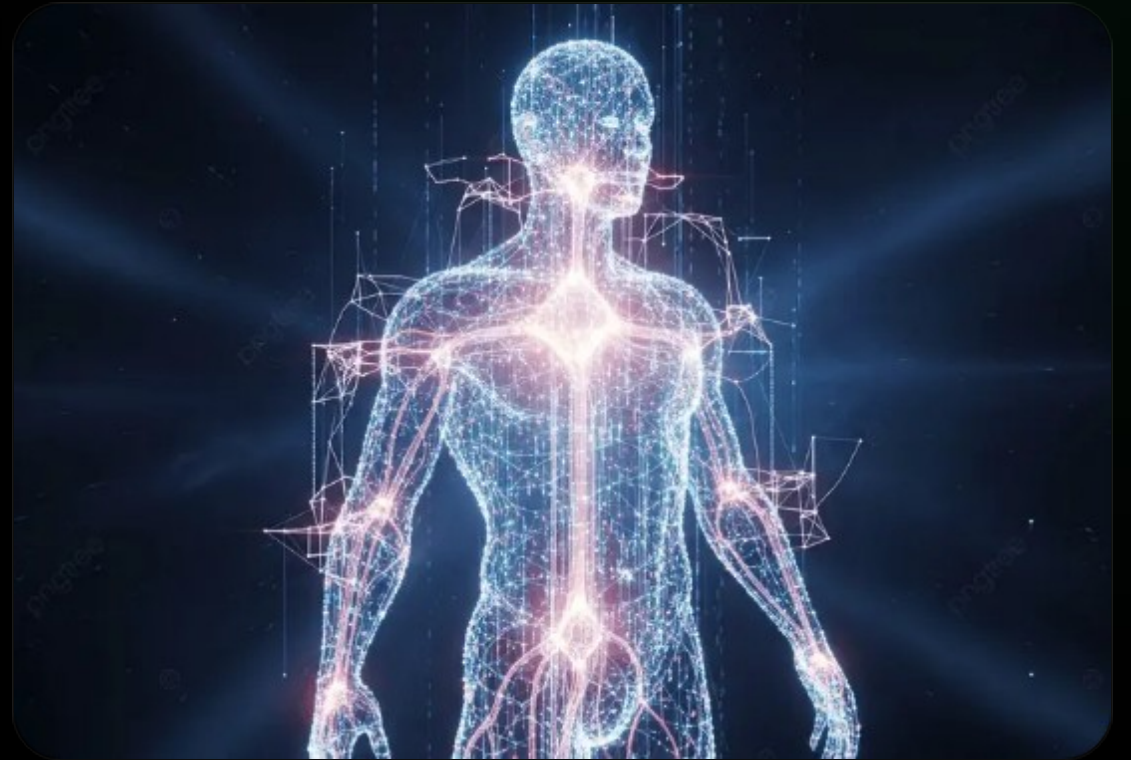
Indoor farms (CEA) spend €2B/year on gas. Escalating carbon taxes (target €200+/tonne) make traditional heating unsustainable for high-density medicinal cultivation.

THE CYBER-PHYSICAL MACRO-CONTROLLER

AI "BRAIN" ORCHESTRATION

INTELLIGENT MACRO-MANAGEMENT

- ✓ **Dynamic Synchronization:** Real-time engine balancing thermal discharge from servers with plant metabolic needs.
- ✓ **Neural Telemetry:** Multi-agent networks continuously adjusting micro-climates and bio-feedback loops.
- ✓ **Autonomous Optimization:** Seamlessly predictive control of fish-to-plant nutrient transfer.



INTEGRATED CIRCULAR ECOSYSTEM

CLOSED-LOOP AQUAPONICS

Symbiotic system converting aquaculture byproduct into ultra-clean bio-nutrients for high-density, organic botanical cultivation.

MEDICINAL FOCUS

Precision indoor climate recipes optimized for pharma-grade botanicals, essential oil potency, and standardized compounds.

THERMAL SYNERGY

Capturing 32-45°C waste heat from nearby AI data centers to eliminate heating costs while ensuring EU compliance.

UNLOCKING MAXIMUM VALUE



PRECISION BOTANICALS

Golden Greens achieves up to **5x higher margins** than basic leafy greens.

Our AI macro-controller maintains millimeter-precise bio-stability, guaranteeing consistent alkaloid and terpene yields for pharmaceutical and wellness buyers.

SYSTEM ARCHITECTURE COMPARISON

FEATURE / CAPABILITY	SMART GREENHOUSE	BASIC VERTICAL FARM	GOLDEN GREENS CORE
Control Layer	Rule-based Timers	Basic IoT Sensors	AI Macro Controller + ML
Nutrient Source	Chemical Fertilizers	Chemical Hydroponics	Circular Aquaponics Bio-Loop
Crop Specialization	Low Margin Vegetables	Leafy / Microgreens	Pharma-Grade Medicinal Herbs
Energy Off-Take	None (Gas Dependent)	High Power Cost	Data Center Thermal Coupling

PRODUCTION OPEX ADVANTAGE

Traditional Indoor Farm

100% OPEX

Hydroponic Greenhouse

70% OPEX

Golden Greens AI Core

30% OPEX (-70%)

The combination of zero-cost thermal integration and nutrient recycling slashes costs by up to 70%.

| AUTONOMOUS IMPACT & YIELD

98%

AI Automation Efficiency

3.5x

Bioactive Potency Increase

95%

Water Loop Circularity

Our predictive cyber-physical control reduces human error and maintains hyper-controlled climate stability 24/7/365.

TECHNOLOGY & SCALING ROADMAP



Q1: AI TRAINING

Testing macro-layer algorithms on aquaponic water chemistry.



Q2: PILOT LOOP

Integration with 1MW data center node co-location.



Q3: HARVEST

First commercial yield of standardized pharmaceutical herbs.



Q4: LICENSING

Deploying "Brain" modules to partner farms across EU.

An aerial photograph of a city, likely San Francisco, showing a mix of modern high-rise buildings and older structures. A body of water, possibly the San Francisco Bay, is visible in the background. The image is overlaid with a dark, semi-transparent layer to make the text stand out.

OUR SHARED VISION

BUILDING THE AI BRAIN FOR CIRCULAR AGRICULTURE

Combining data center energy, aquaponic biology, and precision medicinal chemistry into one unified cyber-physical ecosystem.

CONNECT WITH US

Let's power the next era of smart farming and bio-tech together.

VIKTORS SIMONENKO

FOUNDER & VISIONARY

-  SIA "Golden Greens Crop"
-  Cyber-Physical Core Division
-  Riga, European Union

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